A Covariance-Aware Reanalysis of H_0 Using Public Pantheon+SH0ES Data

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ABSTRACT

Context. In light of the persistent discrepancy between the values for the Hubble constant obtained by local distance-ladder determinations and those obtained by early-Universe inferences using ΛCDM, it is crucial to have robust analyses with well-characterized uncertainties. In particular, correlated statistical and systematic uncertainties must be treated consistently when propagating calibrated Supernova and Cepheid data.

Aims. This work assesses whether a covariance-aware analysis of the public Pantheon+SH0ES products, using the published data, without performing the Cepheid calibration, yields a stable local inference of H_0, and identifies which modeling choices primarily affect its central value or its uncertainty.

Methods. For our analysis, we combine 77 calibrator rows and 277 Hubble-flow rows using the published complete covariance matrix (statistical + systematic). Published Cepheid distance moduli constrain the standardized Type Ia supernova absolute magnitude $M_{\rm SN}$ for the calibrator rows, while a third-order cosmographic distance relation links the Hubble-flow rows to H_0. The two-parameter model is solved analytically with generalized least squares and sampled independently with MCMC. Predefined variants test the distance model, covariance treatment, fixed cosmographic parameters, and sample selection.

Results. The analytic generalized least-squares solution gives $H_0 = 73.529 \pm 1.048\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and $M_{\rm SN} = -19.2469 \pm 0.0299$ mag. MCMC gives $H_0 = 73.55^{+1.06}_{-1.03}\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and $M_{\rm SN} = -19.246^{+0.030}_{-0.030}$ mag, in close agreement. Scientifically reasonable variants shift H_0 by at most $0.162\,{\rm km\,s^{-1}\,Mpc^{-1}}$. A first-order distance law shifts it by $-2.955\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and increases χ^2 by 32.68, whereas discarding off-diagonal covariance terms shifts it by only $-0.033\,{\rm km\,s^{-1}\,Mpc^{-1}}$ but artificially reduces the reported uncertainty by 37.3%.

Conclusions. Within the tested model space, the inferred value of H_0 is stable under scientifically reasonable analysis variants. The robustness tests distinguish two effects: the adequacy of the distance–redshift relation primarily affects the central value, whereas the covariance treatment determines whether the reported uncertainty is reliable. Because the public inputs already incorporate the SH0ES Cepheid calibration, this analysis constitutes a covariance-aware robustness study of calibrated Pantheon+SH0ES products and should not be interpreted as an independent determination of H_0.

Key words. cosmological parameters – distance scale – methods: statistical – supernovae: general

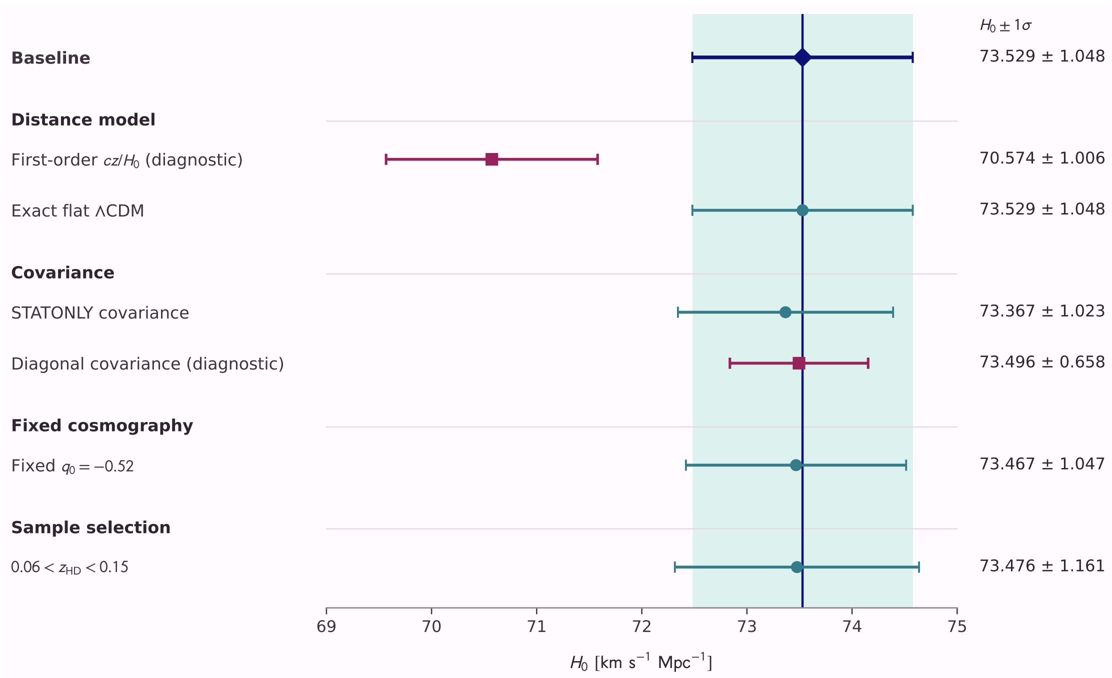

Fig. 1. Internal robustness of the inferred Hubble constant. Horizontal bars show the generalized least-squares 1σ intervals. The variants reuse the same observations and should not be interpreted as independent measurements.

Main analysis: Covariance-aware H_0 reanalysis **Project repository:** Complete code and notebooks
Data and selected references: Pantheon+SH0ES Data Release; Brout et al. (2022); Riess et al. (2022).